

Chapter 4: Logic Gates with Boolean Functions

This chapter explores the fundamental building blocks of digital electronic circuits. Just as we use signals in daily life (like traffic lights) to make decisions, computers use electronic signals processed through logic gates.

4.1 Introduction to Logic Gates

Logic circuits enable the construction of logical conditions using binary values (0 and 1). These circuits are essential for decision-making within the CPU. Logic gates can be classified into two main types:

- **Basic Logic Gates:** AND, OR, and NOT gates.
- **Combinational Logic Gates:** NAND, NOR, XOR, and XNOR gates.

4.2 Basic Logic Gates

1. AND Gate

The output of an AND gate is 1 only if all its inputs are 1. If any input is 0, the output is 0. This is represented by the Boolean expression $Q = A \cdot B$.

Input A	Input B	Output Q ($A \cdot B$)
0	0	0
0	1	0
1	0	0
1	1	1

2. OR Gate

The output of an OR gate is 1 if at least one of its inputs is 1. It is 0 only when all inputs are 0. This is represented by the Boolean expression $Q = A + B$.

Input A	Input B	Output Q (A + B)
0	0	0
0	1	1
1	0	1
1	1	1

3. NOT Gate (Inverter)

A NOT gate has only one input and one output. It reverses the logical state. If the input is 1, the output is 0, and vice versa. This is represented as $Q = \bar{A}$.

Input A	Output Q ($\bar{A}$)
0	1
1	0

4.3 Combinational Logic Gates

These gates are formed by combining basic logic gates with a NOT gate.

1. NAND Gate (NOT + AND)

The output is the inverse of an AND gate. It is 0 only if both inputs are 1. Expression: $Q = A \cdot B$.

Input A	Input B	Output Q
---------	---------	----------

0	0	1
0	1	1
1	0	1
1	1	0

2. NOR Gate (NOT + OR)

The output is the inverse of an OR gate. It is 1 only if both inputs are 0. Expression: **$Q = A + B$** .

Input A	Input B	Output Q
0	0	1
0	1	0
1	0	0
1	1	0

4.4 Integrated Circuits (ICs)

Modern computers do not use individual large gates. Instead, thousands or millions of gates are etched onto a single small silicon chip called an **Integrated Circuit (IC)**. These are identified by specific numbers (e.g., 74 series).

Summary: Every decision in a computer system, from simple arithmetic to complex software logic, is handled through these basic electrical signal patterns. Understanding logic gates is the first step toward understanding computer architecture.